WORKED SOLUTION

STEP II 1995, Question 8

Pure • Differential equations

STEP II 1995, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q8

This question has a full guided treatment in *Beyond STEP*, Section 8.3, *Harvesting a Population*: https://beyondstep.keithmcnulty.org/ch08/harvesting.html

The key idea

The growth law is a logistic equation, which is separable: partial fractions turn the integral into two logarithms. For the second part, completing the square in the new right-hand side shows that the shift $y = x - K + \alpha$ turns the new equation into *exactly the same* logistic equation, with α in place of K. So we can reuse the first answer without doing any more integration.

Solution

The population without removal

We take $K > 0$ (a population that starts at $x = K$ must have $K > 0$). The equation is

$$\frac{\mathrm{d}x}{\mathrm{d}t} = (2K - x)x.$$

Starting from $x = K$, the rate is $K^2 > 0$. The solution can never reach $x = 0$ or $x = 2K$, because those constant functions are themselves solutions and solutions of this equation can't cross. So $0 < x < 2K$ throughout, and we may divide by $(2K - x)x$. In partial fractions,

$$\frac{1}{x(2K - x)} = \frac{1}{2K}\left(\frac{1}{x} + \frac{1}{2K - x}\right),$$

so separating the variables gives

$$\frac{1}{2K}\int\left(\frac{1}{x} + \frac{1}{2K - x}\right)\mathrm{d}x = \int \mathrm{d}t,$$

$$\frac{1}{2K}\Big(\ln x - \ln(2K - x)\Big) = t + C.$$

No modulus signs are needed, since x and $2K - x$ are both positive. At $t = 0$ we have $x = 2K - x = K$, so the left-hand side is 0 and $C = 0$. Hence

$$\ln \frac{x}{2K - x} = 2Kt \quad \Longrightarrow \quad x = (2K - x)\,\mathrm{e}^{2Kt} \quad \Longrightarrow \quad x = \frac{2K\mathrm{e}^{2Kt}}{1 + \mathrm{e}^{2Kt}} = \frac{2K}{1 + \mathrm{e}^{-2Kt}}.$$

To match the required form, write $2 = (1 + \mathrm{e}^{-2Kt}) + (1 - \mathrm{e}^{-2Kt})$:

$$x(t) = \frac{K(1 + \mathrm{e}^{-2Kt}) + K(1 - \mathrm{e}^{-2Kt})}{1 + \mathrm{e}^{-2Kt}} = K + K\,\frac{1 - \mathrm{e}^{-2Kt}}{1 + \mathrm{e}^{-2Kt}},$$

which is the result we needed. (Equivalently, $x = K + K\tanh Kt$.)

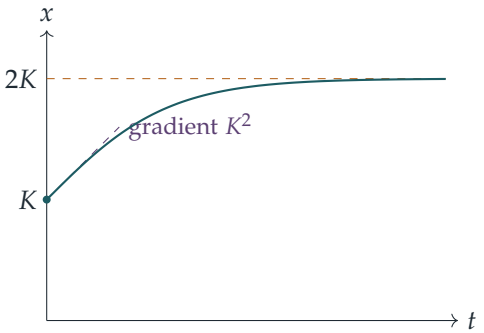
The sketch. For $t \geq 0$ we have $\mathrm{e}^{-2Kt} \in (0, 1]$, so $K \leq x < 2K$. Then $\dot{x} = (2K - x)x > 0$, so x is increasing. Differentiating the equation again,

$$\ddot{x} = (2K - 2x)\,\dot{x} \leq 0 \quad \text{since } x \geq K,$$

so the graph is concave, with gradient K^2 at $t = 0$ (where $\ddot{x} = 0$: the origin of time is the point of inflection of the full S-shaped logistic curve). As $t \to \infty$, $\mathrm{e}^{-2Kt} \to 0$, so

$$x(t) \to 2K.$$

The population levels off at $2K$, approaching it from below and never reaching it.



With removal at rate L

Removing L micrograms per hour subtracts L from the rate of change:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = (2K - x)x - L.$$

Complete the square in the right-hand side:

$$(2K - x)x - L = -(x^2 - 2Kx) - L = K^2 - L - (x - K)^2 = \alpha^2 - (x - K)^2,$$

where $\alpha = \sqrt{K^2 - L} > 0$ because $K^2 > L$.

Ask yourself Why is $y = x - K + \alpha$ the right substitution?

The right-hand side $\alpha^2 - (x-K)^2$ vanishes at $x = K \pm \alpha$, the two equilibrium populations. Shifting by $K - \alpha$ moves the lower equilibrium to 0, which is where the original logistic equation had its lower equilibrium too. With $x - K = y - \alpha$ and $\dot{y} = \dot{x}$,

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \alpha^2 - (y-\alpha)^2 = 2\alpha y - y^2 = (2\alpha - y)\, y.$$

This is the first equation with α in place of K. The initial condition matches as well: $x(0) = K$ gives $y(0) = \alpha$, the analogue of $x(0) = K$. Since $\alpha > 0$, the first part applies word for word, and

$$y(t) = \frac{2\alpha}{1 + \mathrm{e}^{-2\alpha t}}, \qquad \text{so} \qquad x(t) = K - \alpha + \frac{2\alpha}{1 + \mathrm{e}^{-2\alpha t}}.$$

As $t \to \infty$, $\mathrm{e}^{-2\alpha t} \to 0$, so $y \to 2\alpha$ and

$$x(t) \to K - \alpha + 2\alpha = K + \alpha,$$

as required. Where $L > 0$ we have $\alpha < K$, so the harvested population settles at the lower level $K + \alpha < 2K$, as we'd expect.

Checks

- At $t = 0$ both formulae give $x = K$. Setting $L = 0$ gives $\alpha = K$, and the second formula reduces to $2K/(1 + \mathrm{e}^{-2Kt})$, the first.
- Differentiating $x = K + K\tanh Kt$ gives $\dot{x} = K^2 \operatorname{sech}^2 Kt = K^2(1 - \tanh^2 Kt) = (K - K\tanh Kt)(K + K\tanh Kt) = (2K - x)x$.
- Numerically, with $K = 1$, $x(1) = 2/(1 + \mathrm{e}^{-2}) \approx 1.7616$; with $K = 1$ and $L = 0.75$ (so $\alpha = \frac{1}{2}$), $x(1) \approx 1.2311$. Both agree with a direct numerical (Runge–Kutta) solution of the differential equations, and the second settles at $1.5 = K + \alpha$.

Common slips

- Losing the factor $\frac{1}{2K}$ in the partial fractions, which gives e^{-t} in place of e^{-2Kt}.
- In the second part, writing $+L$, or forgetting that the constant of integration must be found again. Here the substitution makes $y(0) = \alpha$, so no new constant is needed.

Beyond STEP

The second part is a first look at *stability of equilibria*. The equation $\dot{x} = \alpha^2 - (x-K)^2$ has equilibria at $K \pm \alpha$: the upper one attracts nearby solutions and the lower one repels them. You can see this from the sign of the right-hand side, without solving

anything. This phase-line approach, and what happens when $L > K^2$ (no equilibria, so the population is wiped out in finite time), are treated in Boyce and DiPrima, *Elementary Differential Equations*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.